

A review of eye-to-eye videoconferencing techniques

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This paper reviews techniques for achieving eye contact between participants in a videoconference, where video screens (more than 30 cm) have forced the video camera out of acceptable alignment with the participant's face. The techniques are of particular use with screens portraying life size images, where camera misalignment becomes unacceptable.

It also addresses the role of eye contact in interactions between two people to demonstrate the value of preserving it.

1. Introduction

In an age of increased communication, with greater availability and use of broadband telecommunications networks, there has been a corresponding expansion in the use of video-based telecommunications.

As a result, video codec equipment is becoming more widely available, at a lower price. One key application of video-based telecommunications is videoconferencing. Videoconferencing enables people who are remotely located from one another to communicate using both sight and sound. In effect, videoconferencing reduces the communications void that exists between traditional audio-based telephone conversations and real-life person-to-person meetings. Figure 1 illustrates a typical videoconferencing display and camera configuration.

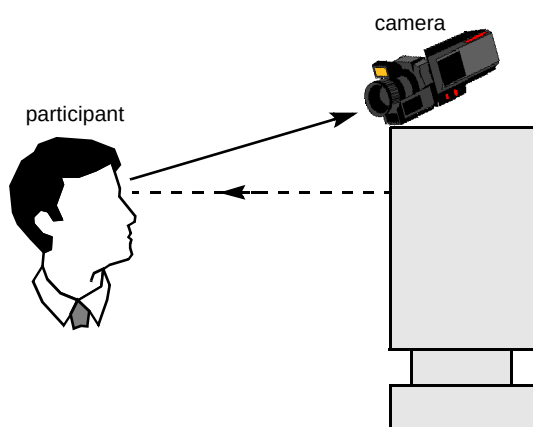


Fig 1 Conventional videoconferencing arrangement.

Although videoconferencing systems are very effective as a means of remote person-to-person communication, they

do not totally fill the need for actual person-to-person meetings, with associated time and travelling costs. In the UK alone, it is estimated that £14 billion is spent on business travel per year.

Several innovative approaches have been described that increase the feeling of presence between participants in a videoconference meeting, in effect increasing the level of 'telepresence'. Telepresence may be considered as the use of technology to establish a sense of shared presence or shared space among geographically separated members of a group. The advancement of telepresence generally requires the integration of telecommunications, computing and audio/video technologies.

One method of increasing the feeling of telepresence in videoconference meetings is to use large screen displays. This results in a more realistic and life-size image of the remote participant, with more easily recognisable body language.

One problem that reduces the level of telepresence during videoconference meetings is the inability of participants to maintain eye contact with each other, with a corresponding loss of rapport. As conventional videoconferencing systems typically place the camera either to one side or above the display screen, the camera sees an off-angle view of the participant and eye contact between participants cannot be achieved. This unwanted effect is further worsened by the increased use of large screen displays, as might be used in the studio.

This paper provides the reader with a basic understanding of eye contact and gives a review of various

proposed solutions. It also describes some of the experimental work carried out at BT Laboratories.

2. What is eye contact ?

Eye contact in this context refers to the sensation of looking into another person's eyes.

The perception of eye contact is very subjective, varying from absolute line of sight between eyes to generally looking in the direction of a person. Therefore it is difficult to measure eye contact in the laboratory [1]. Psychologists often refer to gaze (looking directly at the face of another person) and mutual gaze (two persons looking directly at each other), though these are used interchangeably with eye contact in general usage.

The role of eye contact in communications is more important than the definition. Eye contact performs four key functions [2—4]:

- regulating the flow of conversation — eye contact is used to regulate invitations to start a conversation and cues to speak (interestingly these cues can be interpreted differently by people of different ethnic origin, interrupting the flow of conversation [5]),
- monitoring feedback — the presence of gaze can convey interest, attention and attraction, and likewise lack of gaze or breaking off gaze can convey the opposite (this feedback can unintentionally increase attraction of males to females, though females seem less affected [6, 7]),
- expressing emotion — along with the brows and mouth, the eyes are one of three main elements that convey facial emotion; people have been described as having serious or smiling eyes, and obscuring the eyes for instance behind sunglasses causes tension for the other person [8],
- communicating the nature of the interpersonal relationship — gaze can be used to convey like and dislike, and an intensity of positive or negative feeling; it can also convey rank or status between people — a higher status person often displays less eye contact than their lower status converser, implying less need to persuade [9, 10].

Non-verbal contact contributes much to an interaction — we retain as much as 83% of what we see, yet only 11% of what we hear (according to Karch). How much eye contact contributes is not known, but is certainly significant [11].

3. Eye contact solutions

Various methods have been proposed that are intended to increase the feeling of eye contact between participants during videoconference meetings. These methods may be categorised into the following groups:

- light division, where the camera and display simultaneously share the same optical path,
- space division, where the camera may occupy part of the display area, such as using a pin-hole arrangement,
- time division, where the display screen material is switched rapidly from an optically opaque and translucent state in synchronism with the display and camera equipment,
- image processing, where the facial features that effect eye contact are artificially superimposed to give the impression of eye contact.

3.1 Light division

Half-transparent mirror arrangement

This arrangement is possibly the most conventional way of effecting eye-to-eye contact, one of which is available as a commercial product [12]. Light division is used to achieve the effect of eye contact by arranging the camera so that it simultaneously shares the same light path as the display screen. A half-transparent mirror (HM) is placed in front of the viewed screen at an approximate angle of 45°, as illustrated in Fig 2.

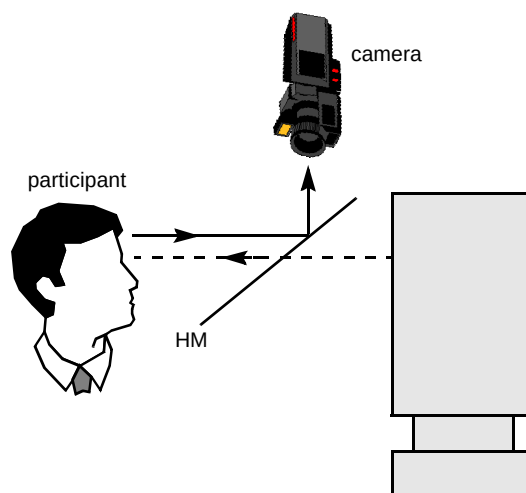


Fig 2 HM videoconferencing configuration.

The HM has the effect of optically changing the axis of the camera. By positioning the camera as shown, the image seen by the camera is the reflected, face-on view of the participant. In practice, a second mirror can be located in the optical path of the camera in order to negate the effects of the single reflection image reversal.

Blazed half-transparent mirror

A development of the HM arrangement, referred to as the blazed half-transparent mirror (BHM) has been described and analysed elsewhere [13, 14]. The name BHM arises due to the structural similarity with a blazed grating.

This technique is based on a novel HM arrangement that can be placed in parallel to the display screen surface. This has the advantage that, while still enabling eye contact, the BHM material does not protrude forward of the display as in the HM arrangement. The enables design of a more compact and less intrusive videoconferencing terminal that enables participants to effectively touch the display surface.

The formation of the novel BHM structure can be considered in stages as illustrated in Fig 3. A conventional HM surface, normally inclined at an angle of 45° , is divided into millimetre sized pieces, referred to as micro-HMs. By moulding many micro-HMs on to a single sheet, a BHM surface is formed.

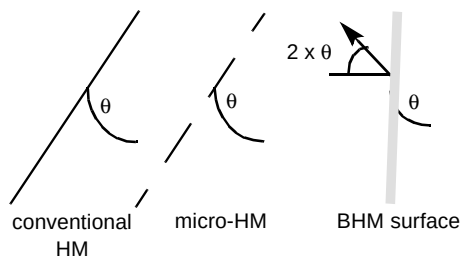


Fig 3 Formation of BHM sheet.

In practice the BHM sheet would be used as a conventional HM but vertically oriented rather than inclined at 45° , as illustrated in Fig 4.

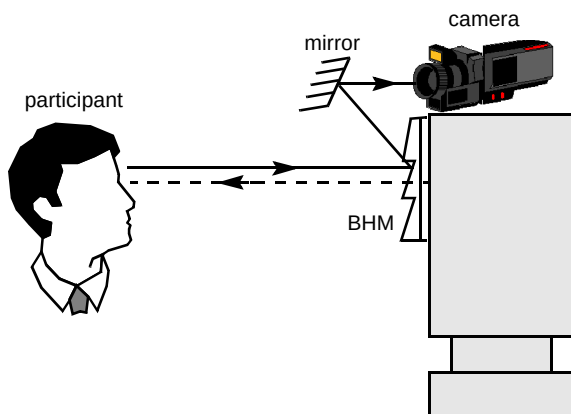


Fig 4 BHM videoconferencing configuration.

3.2 Space division

The space division method for achieving eye contact between the camera and participant uses a miniature camera. The camera is located behind the display surface, at

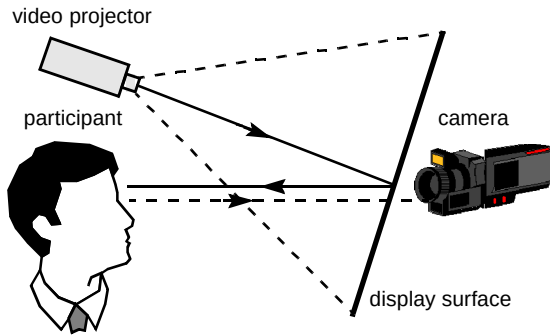


Fig 5 Space division videoconferencing configuration.

a height corresponding to the participant's eye level, as illustrated in Fig 5.

A small hole, the size of the camera lens aperture, is placed in the screen surface to give the camera, located behind the screen, a clear view of the face-on image of the participant. The viewed image is generated by a front-projection video projector. Using this arrangement, some form of control is required to prevent the camera seeing the light output from the video projector. This might be achieved using either a camera arranged such that its field of view does not include the projector, or possibly by using polarised material between the projector and camera, or alternatively using some form of time division control.

A similar system has been described [15], consisting of an overhead video projector, display screen and two miniature cameras mounted behind the screen. A Macintosh Quadra computer is also used to generate graphics information using the data compatible video projector. An additional feature of this system is the use of two directional microphones, placed to either side of the screen. Should there be multiple participants or a single participant located off-centre to the screen, the microphones detect the direction of the speech and hence automatically select the most appropriate camera.

3.3 Time division

This method of achieving eye contact is obtained by arranging the camera to time-share the same optical path as that which the participant uses to view the display screen. This can be achieved by placing the camera and a rear projection video projector behind a switchable liquid crystal display (LCD) panel, as illustrated in Fig 6.

The LCD panel is switched repetitively from an opaque to translucent state in synchronism with the camera and video projector. The camera and video projector are switched alternatively such that when the LCD panel is opaque the video projector is active, and when the LCD panel is translucent the camera is active.

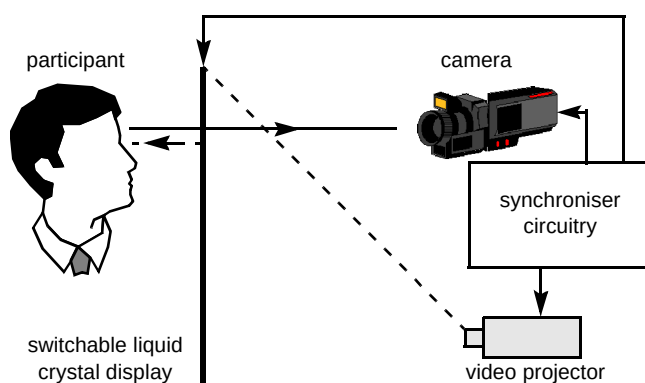


Fig 6 Time division videoconferencing configuration.

This system has an advantage that there is no obtrusive HM or BHM surface placed in front of the display screen, making it especially practical for larger display screen applications. However, the camera and video projector are only active for half of the total time. This results in a 3 dB reduction in the average light level viewed by the camera and similarly a 3 dB reduction in the display illumination intensity. The technique has been described in detail [16—18] using a 40 in switchable LCD panel.

4. Discussion

This review has concentrated only on methods of achieving eye contact in videoconference meetings. It has not considered other techniques for increasing the effect of telepresence. These might include the ability to maintain eye contact between multiple participants [19, 20], provision of 3-dimensional imaging effects [21—25], or the sharing of interactive workspace environments [26—28].

All of the reviewed eye contact methods achieve eye contact without the need for the participants to wear special equipment.

The use of eye contact in videoconference meetings enables the participants to communicate more naturally, even though they are in separate locations.

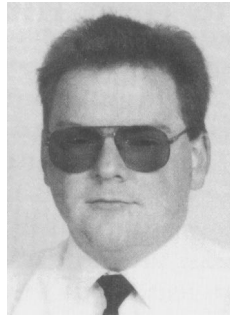
The benefits of using such systems are the reduction in time and costs that would otherwise be incurred travelling to meetings by car, train or plane.

The Advanced Media Unit at BT Laboratories has constructed several videoconferencing terminals that achieve eye contact between participants. The methods used have been based on variations of the HM technique, with added features including a shared interactive workspace. Initial user feedback has been very positive. Other variations are being considered that combine time and space division techniques, using micro-louvered sheets and linear polarised materials.

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Duncan Rose joined BT Laboratories in 1988. He worked initially on data encryption systems, high efficiency modulator and demodulator DSP coding techniques and developed an ultra-high-speed digital equaliser for use in satellite systems.

Since 1993 he has investigated the application of signal processing techniques for the benefit of hearing-impaired telephony users and developed telepresence systems to increase the realism of videoconferencing systems.

He is currently studying for an MSc in Telecommunications Engineering.